

Frozen-DETR: Enhancing DETR with Image Understanding from Frozen Foundation Models

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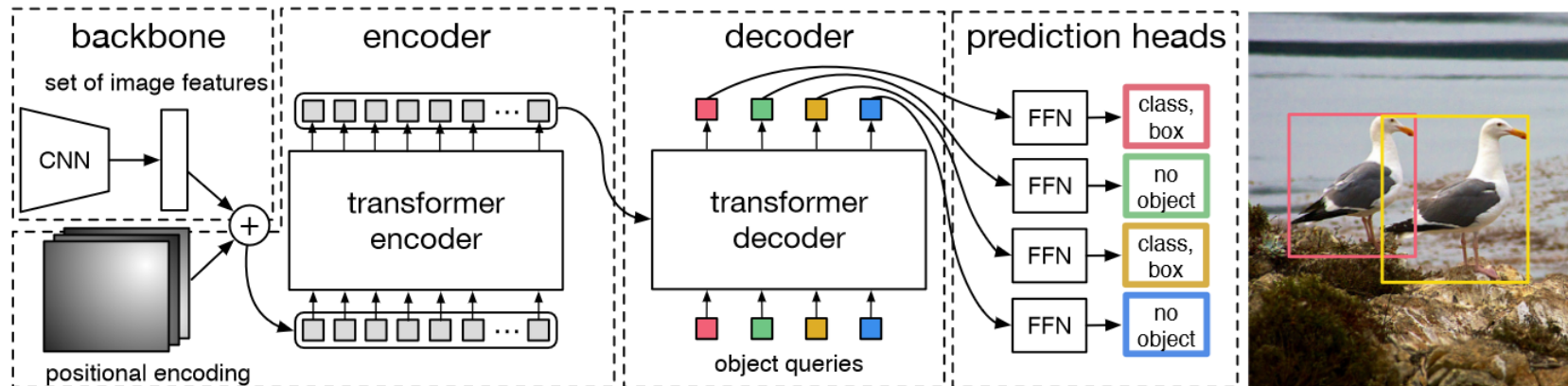
- Problem/Objective
 - Object Detection
- Contribution/Key Idea
 - Propose Frozen-DETR
 - Propose New way to Utilize Foundation Model as a Feature Enhancer
 - Plug-and-Play Module
 - No Foundation Model Fine-tuning Required
 - No Architecture Constraint
 - Asymmetric Input Size
 - SOTA Performance

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- DETR (DEtection TRansformer) [1]

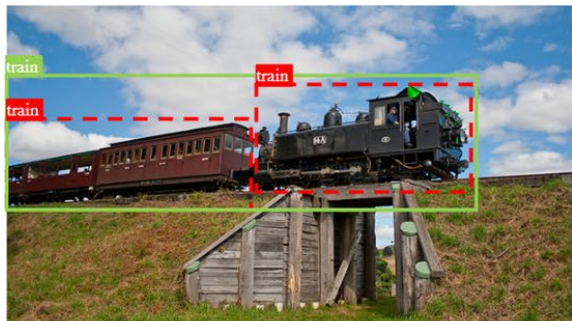


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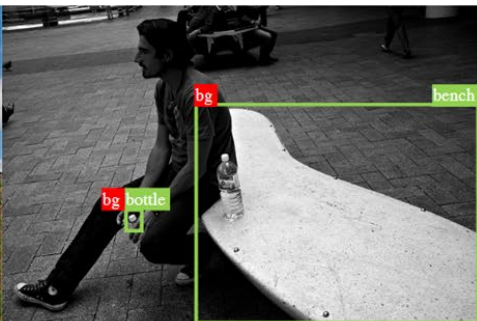
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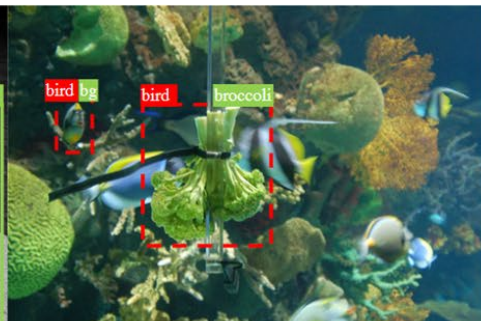
- Global Scene Understanding is Need



(a) clarify the relation between parts and the whole



(b) find missing objects



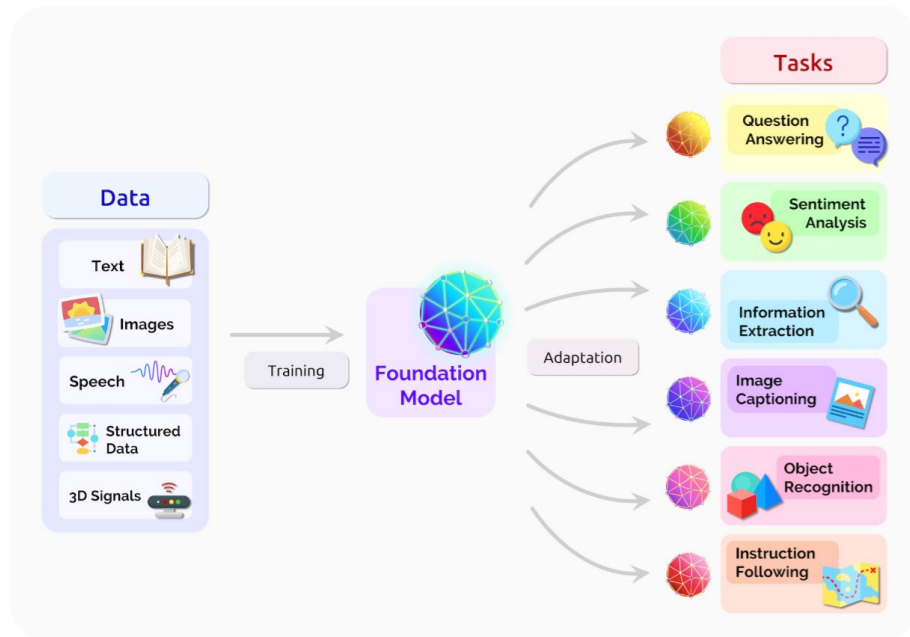
(c) correct wrong classification

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- Visual Foundation Model is Good at Global Context



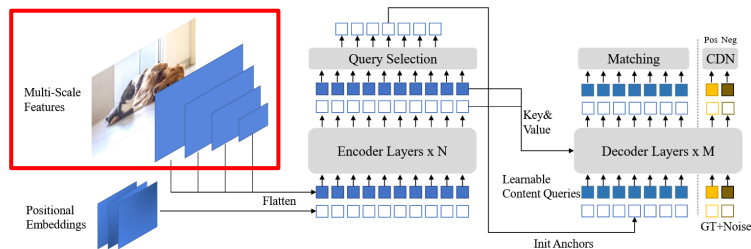
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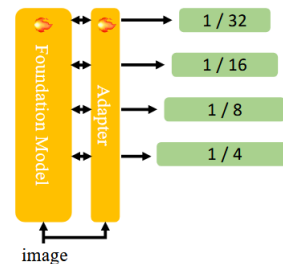
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- Hard to Use Visual Foundation Model as Backbone

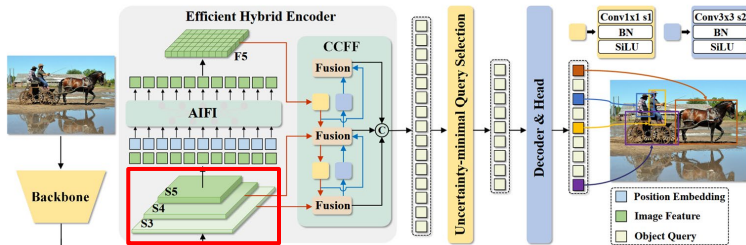
1. Multi-scale feature pyramid



2. Training Computation



3. Object Detection Image Size



896

1333



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- Motivation

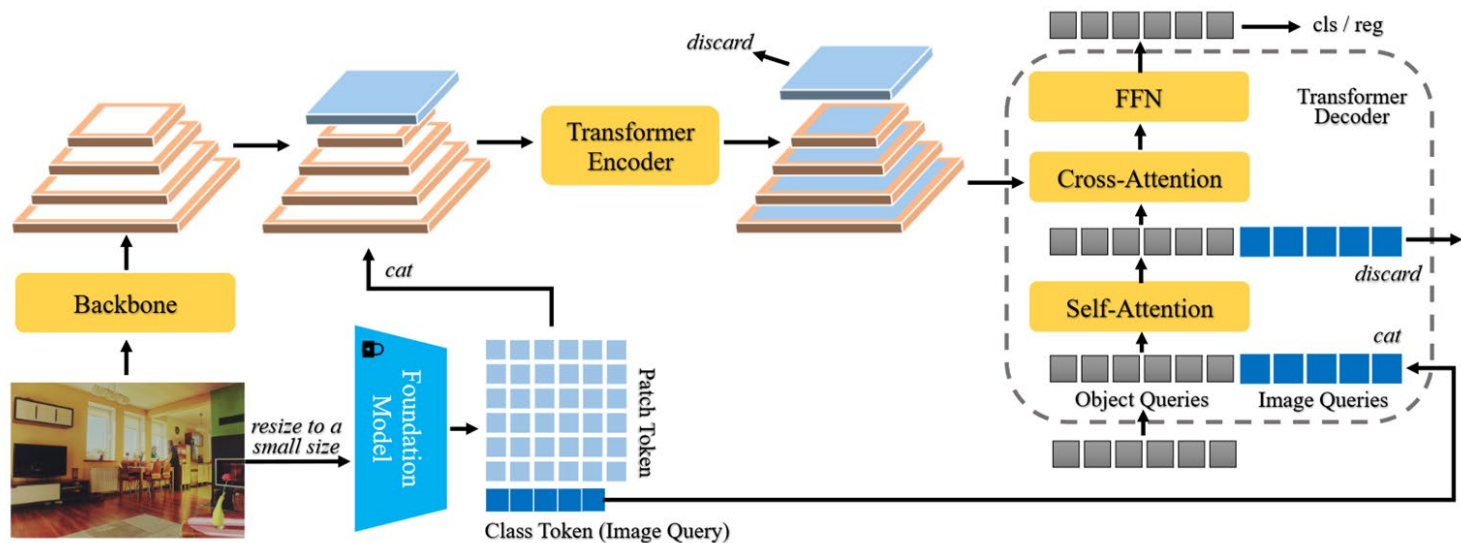
Visual Foundation Model이 가진 Global Scene Understanding 능력을 활용하여 DETR의 성능을 높이면서도 Computation은 합리적인 수준으로 유지할 수 있는 방법이 없을 까?

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Overview

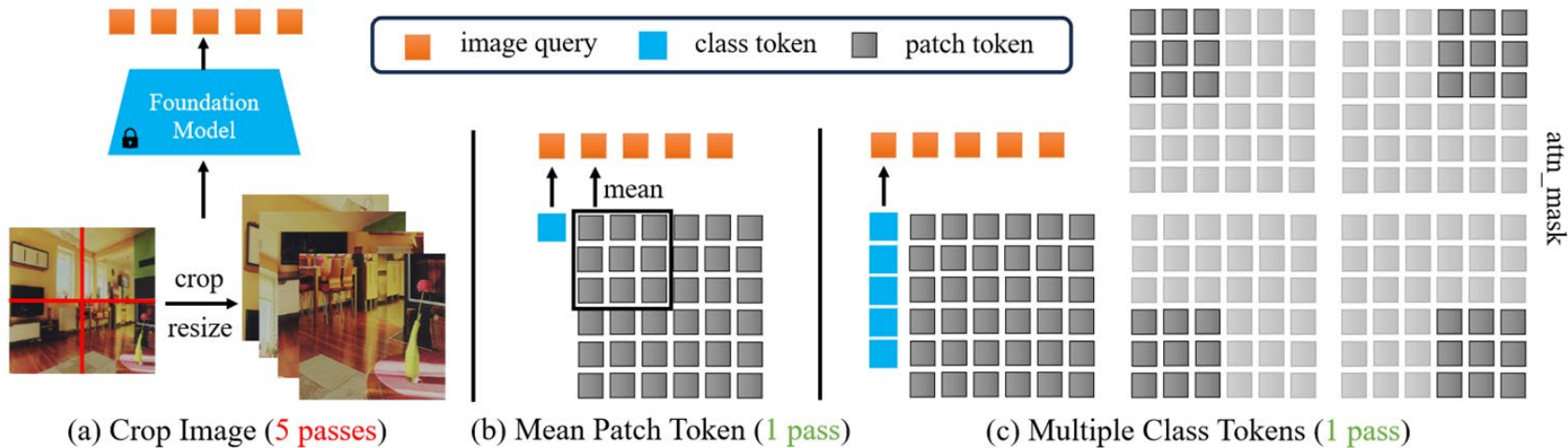


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- Enhancing Decoder by Treating Image Queries

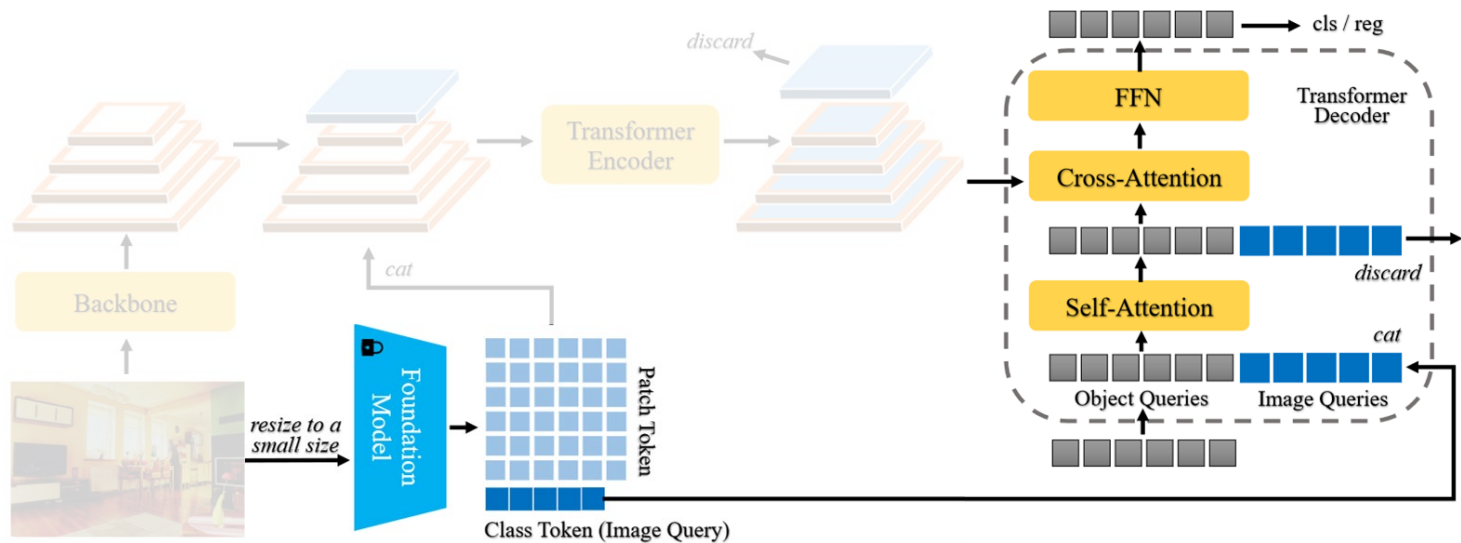


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- Enhancing Decoder by Treating Image Queries

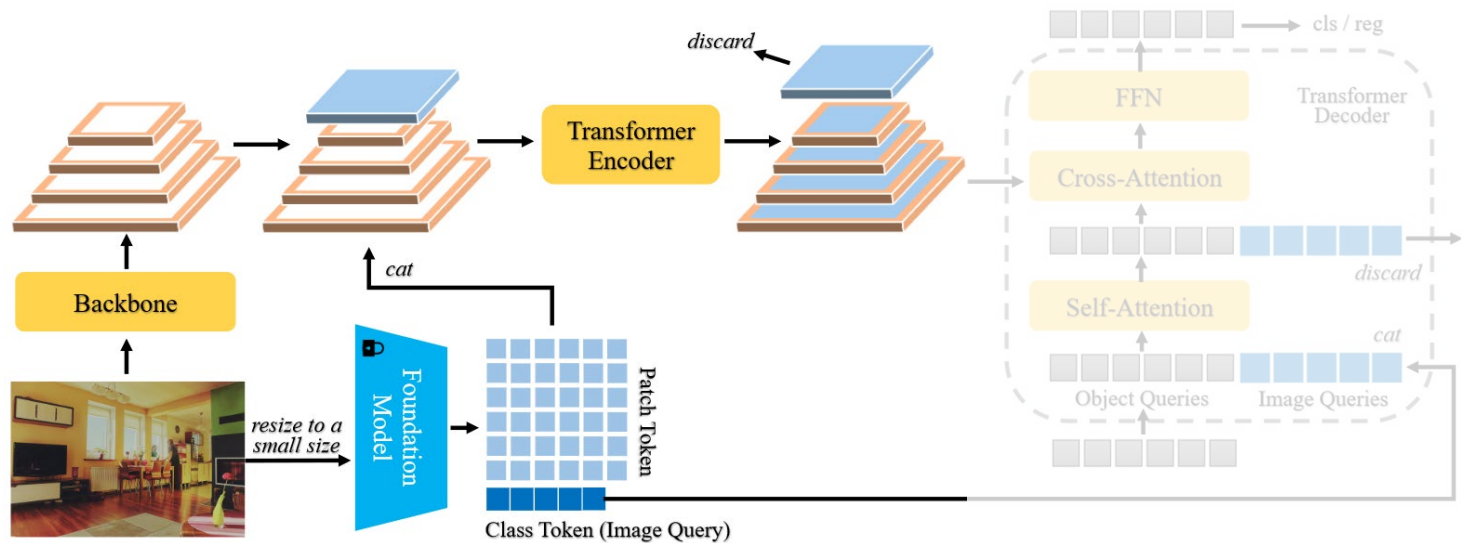


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- Enhancing Encoder by Feature Fusion



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- Experiments

Table 1: Effect of image queries with different detector backbone.

Detector Backbone	Image Query	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
IN-1k R50	✓	44.1	63.3	47.5	26.7	47.0	60.3
IN-1k R50		45.0	64.8	48.7	27.5	47.3	62.1
CLIP R50	✓	45.5	64.9	49.1	28.7	48.8	59.1
CLIP R50		46.1	66.2	49.7	28.8	48.9	61.2

Table 2: Effect of using different foundation models to extract image queries.

Foundation Model	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
N/A	44.1	63.3	47.5	26.7	47.0	60.3
Self	44.2	63.3	47.8	26.4	47.4	60.1
DEiT-III [57]	45.1	64.5	48.5	26.7	47.6	62.1
OpenAI CLIP [54]	45.0	64.8	48.7	27.5	47.3	62.1
DINOv2 [52]	44.6	63.9	48.2	27.6	47.2	60.6
MAE [27]	44.4	63.6	47.7	26.8	47.4	60.3
BEiT-3 [61]	44.1	63.2	47.9	26.7	47.0	60.8

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Experiments

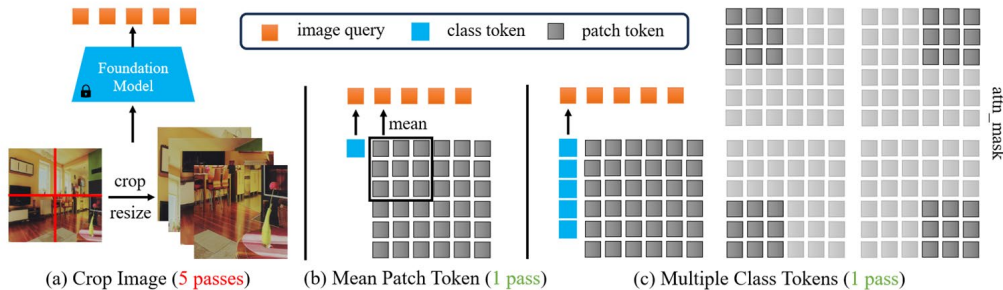


Table 3: Effect of different implementations to extract image queries.

Method	# Queries	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	t_{train}	t_{infer}
Crop Image	1	45.0	64.8	48.7	27.5	47.3	62.1	26h	9.3fps
Crop Image	1+4	45.7	65.7	49.5	28.5	48.8	62.4	44h	3.7fps
Mean Patch Token	1+4	45.3	65.2	49.2	27.7	48.2	61.3	26h	9.1fps
Multiple Class Tokens	1+4	45.8	65.7	49.4	28.7	48.1	63.0	26h	9.2fps

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- Ablation Studies

Table 4: Ablation studies on feature fusion in the encoder.

Exp.	Method	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	Mem	GFLOPs	FPS
1	baseline (no foundation model)	49.0	66.5	53.6	30.6	52.5	64.2	13G	279	9.7
2	Exp. 1 + 5 image queries	50.8	68.8	55.6	33.0	53.9	67.4	14G	392	6.7
3	Exp. 2 + patch tokens to encoder	52.6	70.9	57.4	35.0	56.1	70.4	15G	400	6.5
4	Exp. 3 + patch tokens to decoder	52.7	71.1	57.6	34.5↓	56.0↓	70.8	15G	400	6.5

Table 5: Ablation studies on the input image size of the foundation model.

input image size	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	GFLOPs	FPS
224	51.5	69.7	55.9	32.5	54.6	69.7	333	7.7
336	52.6	70.9	57.4	35.0	56.1	70.4	400	6.5
448	53.1	71.6	58.2	33.9↓	56.6	71.0	494	4.9
560	52.8	71.1	57.5	33.4↓	56.6	70.5	615	3.7

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- Ablation Studies

Table 6: Ablation studies on the model size of the foundation model.

model size	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	GFLOPs	FPS
-	49.0	66.5	53.6	30.6	52.5	64.2	279	9.7
R101 (640)	50.1	68.0	54.9	33.0	53.5	65.7	356	7.6
ViT-B-16 (320)	50.7	68.5	55.5	32.6	53.6	67.8	304	8.4
ViT-L-14 (336)	52.6	70.9	57.4	35.0	56.1	70.4	400	6.5

Table 7: Ablation studies on whether fine-tuning the foundation model (CLIP R101).

Foundation Model	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	Mem
N/A	49.0	66.5	53.6	30.6	52.5	64.2	13G
Trainable	49.0	66.7	53.5	31.1	52.3	64.3	17G
Frozen	50.1	68.0	54.9	33.0	53.5	65.7	14G

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- Experiments

Table 8: Comparisons with other query-based detectors on COCO minival set. *: the input size of the foundation model is 448. †: The single-scale detector uses standard attention in the encoder while Frozen-DETR uses deformable attention to fuse multi-scale features.

Detector	Backbone	# Epochs	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	GFLOPs	FPS
DETR [3]	R50	500	43.3	63.1	45.9	22.5	47.3	61.1	86	27.8
Deformable DETR [73]	R50	50	43.8	62.6	47.7	26.4	47.1	58.0	173	13.4
Sparse R-CNN [56]	R50	36	45.0	63.4	48.2	26.9	47.2	59.5	174	17.8
AdaMixer [24]	R50	36	47.0	66.0	51.1	30.1	50.2	61.8	132	16.6
DDQ DETR 4scale [67]	R50	24	52.0	69.5	57.2	35.2	54.9	65.9	249	8.6
Group DETR (DINO-det-4scale) [8]	R50	36	51.3	-	-	34.7	54.5	65.3	279	9.7
H-Deformable-DETR [34]	R50	36	50.0	68.3	54.4	32.9	52.7	65.3	268	11.0
DAC-DETR [32]	R50	24	51.2	68.9	56.0	34.0	54.6	65.4	279	9.7
DAB-DETR-DC5 [45]†	R50	12	38.0	60.3	39.8	19.2	40.9	55.4	220	10.2
Frozen-DETR (DAB-DETR-DC5)	R50	12	42.0	63.2	44.9	22.4	45.4	61.1	372	8.5
DN-DETR-DC5 [36]†	R50	12	41.7	61.4	44.1	21.2	45.0	60.2	220	10.2
Frozen-DETR (DN-DETR-DC5)	R50	12	44.4	64.8	47.7	23.8	47.7	64.6	372	8.5
DINO-det-4scale [66]	R50	12	49.0	66.6	53.5	32.0	52.3	63.0	279	9.7
DINO-det-4scale [66]	R50	24	50.4	68.3	54.8	33.3	53.7	64.8	279	9.7
DINO-det-5scale [66]	R50	24	51.3	69.1	56.0	34.5	54.2	65.8	860	4.4
Frozen-DETR (DINO-det-4scale)	R50	12	51.9	70.4	56.7	33.8	54.9	69.3	400	6.5
Frozen-DETR (DINO-det-4scale)	R50	24	53.2	71.8	58.0	35.1	56.5	70.6	400	6.5
MS-DETR [68]	R50	12	50.0	67.3	54.4	31.6	53.2	64.0	252	10.8
Frozen-DETR (MS-DETR)	R50	12	53.0	71.5	57.8	35.1	55.8	70.8	423	6.9
DDQ with HPR [69]	R50	12	52.4	69.9	57.5	35.9	55.5	66.7	283	6.5
Frozen-DETR (DDQ with HPR)	R50	12	55.7	73.9	61.3	38.4	58.8	72.3	467	5.2
Co-DINO-5scale [74]	R50	12	52.1	69.4	57.1	35.4	55.4	65.8	860	4.4
Co-DINO-4scale [74]	Swin-B(22k)	12	56.8	74.9	62.5	41.7	60.9	72.8	513	6.2
Frozen-DETR (Co-DINO-4scale)	R50	12	54.0	72.4	59.1	36.0	58.0	71.5	400	6.5
Frozen-DETR (Co-DINO-4scale)	R50	24	54.3	72.9	59.2	36.6	58.0	72.1	400	6.5
Frozen-DETR (Co-DINO-4scale)*	Swin-B(22k)	12	57.6	75.8	63.2	41.4	61.7	74.3	732	3.8

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- Experiments

Table 9: Results on LVIS v1 training with full annotations. *: Our implementation.

Method	# Epochs	Backbone	AP	AP _r	AP _c	AP _f
Deformable-DETR [34]	24	R50	32.2	20.9	31.1	38.4
Detic (Deformable-DETR) [71]	96	R50	32.5	26.2	31.3	36.6
H-Deformable-DETR [34]	24	R50	33.6	22.2	32.4	39.9
DINO-det-4scale*	24	R50	34.4	22.5	33.4	40.8
Frozen-DETR (Ours)	24	R50	41.0	31.2	41.1	45.1

Table 10: Results on open-vocabulary LVIS. *: Our implementation.

Method	#epochs	backbone	AP	AP _r	AP _c	AP _f
ViLD [25]	460	R50	27.8	16.7	26.5	34.2
DetPro [20]	20	R50	28.4	20.8	27.8	32.4
VLDet [41]	96	R50	33.4	22.9	32.8	38.7
BARON [63]	24	R50	29.5	23.2	29.3	32.5
DK-DETR [38]	70	R50	33.5	22.2	32.0	40.2
DINO-det-4scale*	24	R50	32.5	15.2	32.8	39.8
Frozen-DETR (Ours)	24	R50	40.0	24.0	41.8	44.9

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- Experiments

Table 11: Results of combining multiple foundation models on COCO.

CLIP [54]	DINOv2 [52]	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
		49.0	66.6	53.5	32.0	52.3	63.0
✓		51.9 (+2.9)	70.4	56.7	33.8	54.9	69.3
✓	✓	53.8 (+4.8)	72.3	58.7	35.2	57.5	72.3

Table 12: Results on the COCO-O dataset. The models are trained on the COCO datasets and directly tested on the COCO-O dataset without finetuning. ER denotes Effective Robustness.

Method	Backbone	COCO mAP	COCO-O (mAP)							ER
			Sketch	Weather	Cartoon	Painting	Tattoo	Handmake	Avg.	
DINO-det-5scale [66]	Swin-L	58.5	-	-	-	-	-	-	42.1	+15.76
ViTDet [40]	ViT-H	58.7	-	-	-	-	-	-	34.3	+7.89
DETR [3]	R50	42.0	9.0	30.0	12.3	23.9	11.6	15.7	17.1	-1.82
Deformable DETR [73]	R50	44.5	10.5	30.2	15.1	26.2	10.6	18.6	18.5	-1.49
DINO-det-4scale [66]	R50	49.0	13.8	36.3	18.5	30.7	13.3	22.4	22.5	+0.45
Frozen-DETR (DINO-det+CLIP)	R50	51.9	50.3	46.3	51.4	54.9	52.1	46.0	50.2	+26.8
Frozen-DETR (DINO-det+CLIP+DINOv2)	R50	53.8	52.8	49.3	53.5	56.6	57.7	52.3	53.7	+29.49

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- Experiments

Table 13: Error analysis of models with and without foundation models on COCO.

Method	AP	AP ₅₀ (acc)↑	Loc↓	Cls↓	BG↓	FN↓
DINO-det-4scale	49.0	66.7	7.0	9.6	12.4	4.3
+CLIP	51.9	70.4	7.4 (+0.4)	8.2 (-1.4)	10.5 (-1.9)	3.5 (-0.8)
+CLIP+DINOv2	53.8	72.3	7.1 (+0.1)	7.7 (-1.9)	9.9 (-2.5)	3.0 (-1.3)

Table 14: Comparisons with different methods to improve the performance.

Method	Training		Inference		GFLOPs
	Mem	time / epoch	Mem	FPS	
DINO-det-4scale baseline	13G (bs=2)	1.3h	3G	9.7	279
Frozen-DETR (DINO-det-4scale)	15G (bs=2)	1.4h	3G	6.5	400
DINO-det-5scale	34G (bs=2)	2.6h	5G	4.4	860
DINO-det-4scale + ViT-L backbone	44G (bs=1)	4.2h	10G	2.1	1244

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- Visualization

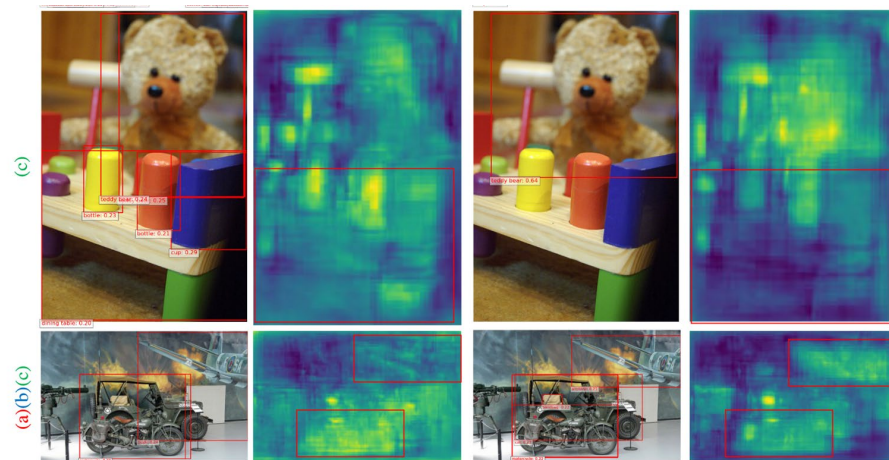
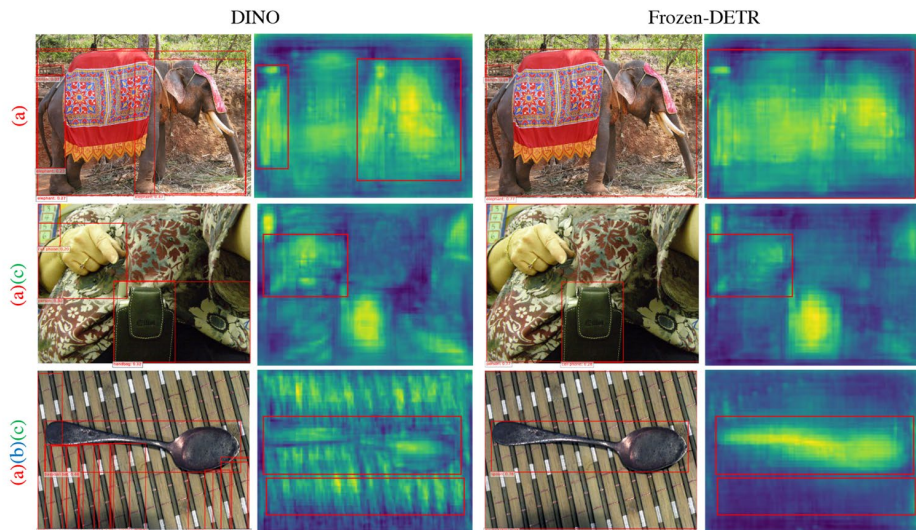


Figure 6: More visualization of the predictions and the feature maps from DINO-det-4scale [66] and Frozen-DETR (CLIP only). Using foundation models can (a) clarify the relation between parts and the whole object, (b) find missing objects, and (c) correct wrong classifications.